PROJECT-I (CEUP 301) — CAPSTONE PROJECT ACADEMIC PRESENTATION

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERI

CLIAS

ADAPTIQ ENGINE

CHARUSAT Learning Intelligence & Assessment System

Longitudinal Knowledge Tracing, Adaptive Practice, In-Browser Coding Sandbox & AI-Proctored Exam Integrity

GROUP ID: CS26029 B.

STUDENT PRESENTERS

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Prof.
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- Bayesian Knowledge Tracing (BKT)
- In-Browser Monaco IDE Sandbox
- AI Integrity Trust Scoring (78.5%)
- Socratic Gemini AI Remediation
- Automated NB Attainment

01 MOTIVATION & ACADEMIC PROBLEM

1. Static Marks Mask True Learning Gaps

A grade like 14/20 provides no visibility into whether correct answers were guessed or wrong answers were slips.

2. No Longitudinal Memory Retention

Concepts decay over time (Ebbinghaus forgetting curve), yet traditional LMS treat past assessments as permanent.

3. Lab Coding Collusion & Accreditation Lag

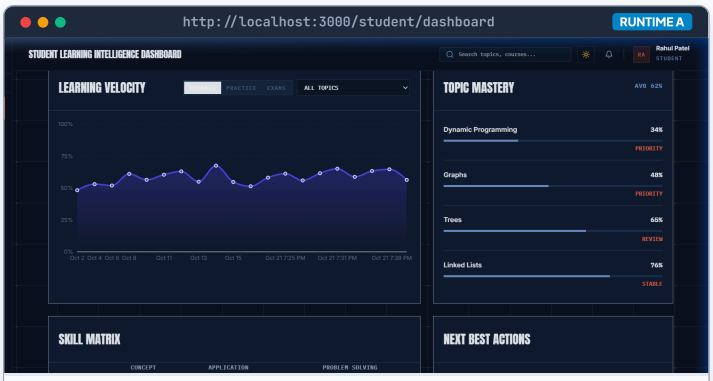
Exams lack live focus tracking, and mapping student work to NBA/NAAC Course Outcomes (CO-PO) is manual.

The Solution: AdaptIQ Core Engine

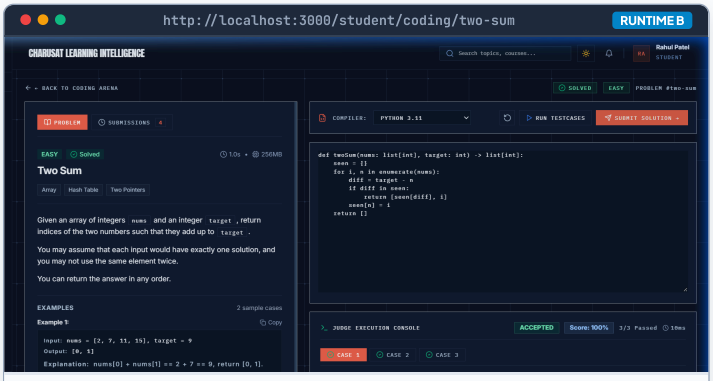
Continuous probabilistic knowledge tracing, real-time code sandboxing, Socratic guidance, and automated accreditation.

LIVE PRODUCTION RUNTIME VERIFICATION

REAL SYSTEM EXECUTION



1. Student Learning Intelligence Dashboard — Recharts Velocity Curve | Topic Mastery | Skill Matrix



2. In-Browser Monaco Coding Arena & Automated Judge — Accepted (100% Score) | 3/3 Testcases | AST Winnowing

02 5-TIER SYSTEM ARCHITECTURE

TIER 1 • CLIENT	Next.js 14 App Router Monaco IDE • Recharts Velocity • Tailwind CSS
TIER 2 • GATEWAY	NestJS Core Gateway TypeScript • JWT Auth • RBAC • SSE Streams
TIER 3 • INTELLIGENCE	FastAPI AI Engine Google Gemini 1.5 • BKT Estimator • IRT Calibration
TIER 4 • SANDBOX	Code Judge & Plagiarism Isolated Subprocess Runner • AST Winnowing
TIER 5 • DATABASE	Persistence & Caching MySQL 8.0 (Prisma ORM) • Redis 7 (BullMQ)

Loop: Diagnostic Drill → BKT Tracing → Code Judge → Socratic Hint → OBE Attainment

03 MATHEMATICAL KNOWLEDGE MODELING

04 SOCRATIC AI

- **Zero-Leak Socratic Tutc** as an encouraging tutor providing direct code or
- **Live Focus Telemetry:** Se capture tab switching an precision.
- **Structural Plagiarism En** prevents renaming and c winnowing.

05 EXPERIMENTA BENCHMARKS

10 ms
CODE SANDBOX LATENC

78.5%
CALIBRATED TRUST SCOR

06 INSTITUTIONA

- **Automated CO-PO Atta** between student proble reports.
- **Counsellor Early-Warnir** automatically based on intervention.
- **Continuous Student Por** capstone defense & emp

BAYESIAN KNOWLEDGE TRACING (BKT)

$$P(L_t) = P(L_{t-1}|Obs) + [1 - P(L_{t-1}|Obs)] \times P(T)$$

Updates latent mastery P(L) based on Slip P(S) and Guess P(G) probabilities after every student submission.

MEMORY DECAY & TRUST INDEX

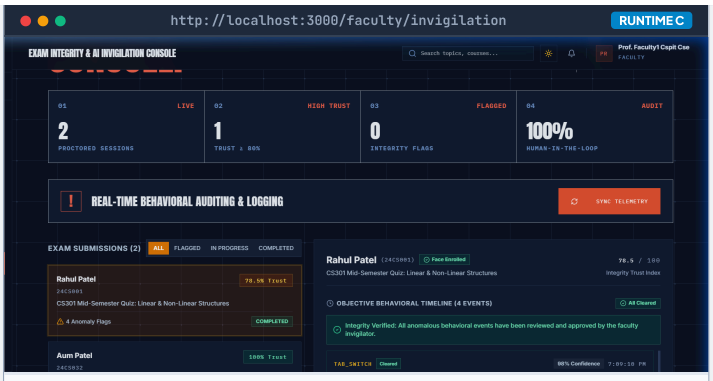
$$Mastery(t) = P(L_0) \times e^{-\lambda \Delta t} \quad | \quad Trust = 100 - \Sigma(w_i \times v_i)$$

Models Ebbinghaus memory decay over inactive days, while exam trust score deducts for window blur and tab switching.

AST WINNOWING PLAGIARISM CHECK

$$Sim(A, B) = |W(A) \cap W(B)| \ / \ |W(A) \cup W(B)|$$

Fingerprints normalized Abstract Syntax Trees against student collusions.



3. Real-Time AI Exam Invigilation & Integrity Guard

78.5% Trust Score

Tab Switch Detection

Window Blur Audit

07 TECHNOLOGY

FRONTEND	Next.js 14
	Tailwind CSS
BACKEND	NestJS (TS)
	AST Winnow
DATABASE	MySQL 8.0
	Docker